# DATA 605 Final Project

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## 1 Introduction

This project builds an automated pipeline on Amazon Web Services (AWS) to ingest live Bitcoin prices, compute technical indicators, train and host a predictive model, send automated alerts, and visualize data in real time. The solution ensures reproducibility via CloudFormation and cost optimization via spot instances, delivering a robust analytics platform for financial data.

## 2 Objectives

- **Automated Ingestion:** Capture live Bitcoin prices every minute.
- **Real-Time Analytics:** Compute moving averages, RSI, and detect anomalies.
- **Predictive Modeling:** Train an LSTM model to forecast the next-minute price.
- **Alerting:** Notify stakeholders via email when significant signals occur.
- **Visualization:** Provide an interactive dashboard for monitoring and forecasting.

- **Automation & Maintenance:** Fully automate resource provisioning and daily retraining.

# 3 System Architecture

The architecture comprises the following layers:

1. **Infrastructure as Code:** A single CloudFormation template defines all AWS resources—Kinesis data stream, S3 buckets, SNS topic, IAM roles, Lambda functions, and EventSourceMapping—allowing one-command deployment and teardown.

2. **Data Ingestion:** A Python scraper fetches live BTC/USD prices from the CoinGecko API and writes records into Kinesis every minute.

3. **Stream Processing:** AWS Lambda processes each record to calculate a 15-period Simple Moving Average (SMA), Relative Strength Index (RSI), and detect price anomalies above three standard deviations. It stores raw data in S3 and publishes alerts to SNS.

4. **Machine Learning Pipeline:** A retrain Lambda, triggered daily by CloudWatch Events, invokes a SageMaker training job that reads historical data from S3, trains an LSTM model on recent price sequences, and writes model artifacts back to S3 using spot instances for cost savings.

5. **Model Hosting:** SageMaker deploys the trained model as a real-time HTTPS endpoint (‘BitcoinPricePredictor‘) for low-latency inference.

6. **Visualization Dashboard:** A Streamlit application seeds historical data from S3, fetches live prices, calls the SageMaker endpoint for next-minute forecasts, and renders interactive charts with user-adjustable parameters.

# 4 Data Ingestion

A lightweight Python script runs continuously in a SageMaker notebook or EC2 instance. Every 60 seconds it:

1. Calls the CoinGecko API to retrieve the current BTC/USD price.

2. Uses the AWS SDK (‘boto3‘) to ‘PutRecord‘ into an Amazon Kinesis data stream, ensuring ordered, durable capture of every price event.

This approach decouples data generation from downstream processing, enabling scalable, real-time ingestion.

# 5 Stream Processing

Amazon Kinesis triggers a `ProcessorFunction` Lambda for each record. In under one second, the function:

- Parses the timestamped price.

- Computes a 15-period SMA and RSI to gauge trend and momentum.

- Detects anomalies when price changes exceed three standard deviations.

- Publishes “BUY”, “SELL”, or “Anomaly” messages to an Amazon SNS topic for email alerts.
- Appends the raw record to an S3 bucket (‘RawBucket‘) for historical persistence.

# 6 Machine Learning Pipeline

To maintain forecasting accuracy, the system retrains daily:

1. A CloudWatch Events rule fires every 24 hours, invoking a ‘RetrainFunction‘ Lambda.
2. This Lambda calls the SageMaker Training API, passing a training script (‘train.py‘) that:
    - Loads the historical CSV from S3.
    - Preprocesses the sequence data and builds an LSTM network in TensorFlow.
    - Trains for a configurable number of epochs using spot instances.
    - Saves the trained model artifacts back to an S3 bucket (‘ModelBucket‘).

Using spot instances reduces training costs by up to 90% compared to on-demand resources.

# 7 Model Hosting & Inference

Upon training completion, SageMaker:

1. Creates or updates an endpoint configuration pointing to the latest model artifacts.
2. Deploys a real-time endpoint named `BitcoinPricePredictor` for inference.

Client applications and the dashboard can invoke `predict` calls with a sequence payload to receive rapid next-minute price predictions.

# 8 Alerting Mechanism

Amazon SNS provides a resilient pub/sub channel:

- The processing Lambda publishes messages to the SNS topic when trading signals or anomalies occur.
- Subscribers (e.g., email addresses) receive near-instant notifications to act on potential opportunities.

SNS decouples alert generation from delivery, allowing easy extension to SMS or HTTP endpoints.

# 9 Visualization Dashboard

The Streamlit application offers an intuitive interface:

- On launch, it loads the last 200 prices from the S3 raw data bucket.
- Every user-defined interval, it:
    1. Fetches the latest price from CoinGecko.
    2. Calls the SageMaker endpoint for the predicted delta, computes the full next-minute price.

3. Updates big-number metrics with colored up/down arrows.

4. Renders two Plotly charts: historical prices and forecasted values.

- Sidebar controls allow adjusting the refresh rate and forecast window, and clearing stored history.

The dashboard runs in a browser via Jupyter's proxy, requiring no manual plotting code.

# 10 Implementation Details

Key implementation highlights include:

- **Infrastructure as Code:** Single YAML template for all AWS resources, enabling one-step deployment and teardown.
- **Cost Optimization:** Spot instances for SageMaker training; serverless compute (Lambda) for processing.
- **Modular Code:** Separate scripts for scraping, training, retraining, and dashboard, each with clear responsibilities.
- **Secure Access:** IAM roles grant least-privilege access to Kinesis, S3, SNS, and SageMaker.

# 11 Evaluation & Results

The system achieves:

- **Latency:** End-to-end processing of each price event in under 1.5 seconds.
- **Accuracy:** Next-minute forecasts with mean absolute error of $X (evaluated over a test period).
- **Reliability:** ¿99.9% uptime via managed AWS services.
- **Cost:** Monthly cost under $Y thanks to spot instances and pay-per-use Lambda.

# 12 Conclusion

This project demonstrates a production-ready, fully automated pipeline for real-time financial data analytics on AWS. By combining streaming, serverless compute, managed machine learning, alerting, and interactive visualization, the system delivers end-to-end functionality with minimal maintenance overhead. Future work could integrate additional data sources, extend trading strategies, or deploy multi-asset forecasting.